

Data Quality Report — BOAMP Renewal Modeling Dataset

Gigalis Internship — Phase 1 Deliverable L1

Internship Project — Gigalis

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1 Objective and Scope

This report documents the quality of the BOAMP corpus used to construct the renewal proxy and the survival modeling dataset, and whether the dataset is reliable enough to support Phase 2. The scope is the BOAMP-only pipeline for digital-sector public procurement notices in Pays de la Loire over 2015–2024. The official modeling population is the set of eligible `APPEL_OFFRE` contracts retained in `data/processed/boamp_phase2_survival.csv`. In this report, `event = 1` denotes a *proxy renewal event* found by the linking algorithm; `event = 0` denotes right-censoring or no observed link under the rule before the study end date. It does not prove that no renewal occurred. The report covers: the raw corpus and final population, missing-value and cleaning decisions, the dominant data-quality issues, duration reliability, buyer identification and SIREN enrichment, the exact construction of the `event` variable, its validation and robustness, and the limitations that carry into survival analysis.

2 Raw BOAMP Corpus

The cleaned BOAMP corpus is stored in `data/processed/boamp_full_clean.csv`. It contains 3,181 notices published between 2015-03-02 and 2024-12-29.

Table 1: Raw BOAMP notice types

Notice type	Count	Share
APPEL_OFFRE	1,933	60.8%
ATTRIBUTION	1,081	34.0%
RECTIFICATIF	150	4.7%
PRE-INFORMATION	7	0.2%
MODIFICATION	5	0.2%
INTENTION_CONCLURE	4	0.1%
PERIODIQUE	1	0.0%
Total	3,181	100.0%

The renewal-linking pipeline uses only the 1,933 `APPEL_OFFRE` notices as possible source contracts. `ATTRIBUTION` notices are used as an auxiliary source to refine contract start dates when a BOAMP back-reference is available.

3 Final Modeling Population

The study population is narrower than the raw corpus because not every notice is a candidate source contract and not every source contract has an observable renewal window before the 2024-12-31 study end.

Table 2: Population construction summary

Population	Count	Interpretation
Raw BOAMP notices	3,181	Full digital-sector corpus downloaded from the BOAMP API.
APPEL_OFFRE source notices	1,933	Potential source contracts for renewal linking.
Eligible source contracts	1,210	APPEL_OFFRE contracts whose estimated end date falls before study end ($W=6$ months).
Linked proxy-renewal events	665	<code>event</code> = 1; a plausible later BOAMP notice was linked.
Right-censored contracts	545	<code>event</code> = 0; no observed link under the rule before study end.

The linking rate in the eligible population is 55.0% (665/1,210). This is the denominator that matters for Phase 2 survival modeling. The lexical baseline retained in the repository for comparison is 7.6% at the same $W = 6$ window (11.3% at the earlier $W = 12$ setting).

4 Missing Values and Cleaning Decisions

Table 3 summarizes the main variable-level quality issues that carry forward into modeling.

Table 3: Variable quality and cleaning decisions

Variable	Issue	Cleaning / treatment	Modeling impact
buyer_siret / buyer_key	Raw buyer_siret coverage is 9.1% in the 3,181-row corpus; post SIREN enrichment, 724 of the 1,210 eligible contracts carry a SIREN: key (plus 2 SIRET:) and 484 a NAME: key.	Build buyer_key as SIREN:... when matched with HIGH confidence via the API Recherche d’Entreprises, else NAME:...	Buyer-name fragmentation is the main source of false negatives in linking; SIREN enrichment partially mitigates this.
CPV fields	cpv_principal is missing for 1.9% of the raw corpus; generic codes reduce specificity. Eligible cpv_div2 coverage is 95.3%.	Clean and standardize CPV; generic codes are capped at division-level similarity; missing CPV is excluded from the composite score with weight renormalization.	Missing or generic CPV weakens auditability and may reduce match discrimination.
amount_clean	Valid amount coverage is 41.1% in the full corpus and 21.4% among APPEL_OFFRE ; 20.3% in the eligible handoff.	Flag zero, tiny, and ceiling values; set flagged amounts to missing.	Amount is too sparse to act as a linking signal and must be handled as optional context only.
declared_duration_months duration_clean	Raw cleaned duration coverage is 47.3% overall and 76.6% for APPEL_OFFRE . In the eligible cohort, 24.4% are imputed.	Remove suspect durations and impute missing eligible durations to 48 months with dur_was_imputed .	Duration affects eligibility and temporal search windows, so imputed rows are less reliable.
start_date	No missing values in the eligible handoff, but 59.3% use publication date and 40.7% use an attribution-derived award date.	Use award date via annonce_lie when available, else publication date.	Temporal scores contain date noise because publication is not always the operational contract start.
type_procedure	0.3% missing in the eligible handoff.	Retain as-is; no imputation required.	Low risk for descriptive summaries and covariate use.
type_marche	14.4% missing in the eligible handoff.	Retain missing values explicitly.	Missingness may blur segment-level interpretation.
event / renewal link	No explicit BOAMP renewal identifier exists; event is algorithmic rather than observed ground truth.	Use the linking algorithm to build a proxy event; keep confidence diagnostics.	False positives and false negatives remain possible, so event quality must be interpreted as bounded evidence.

5 Key Data Quality Issues

The dominant data quality problems are structural rather than cosmetic. Buyer identity was initially weak because the eligible cohort had no SIRET-anchored `buyer_key`; most grouping relied on normalized buyer names. SIREN enrichment (completed 2026-06-24 via the API Recherche d’Entreprises) has partially resolved this: 724 of the 1,210 eligible rows now carry a HIGH-confidence SIREN-enriched buyer identity in `boamp_phase2_survival_siren_enriched.csv`; 484 remain on the name-based key (see Section 7). Residual name-based fragmentation remains possible for the unresolved cases. CPV coverage is high but not perfect, and generic CPV values reduce matching precision. Contract amount is sparse on source notices, so it cannot support renewal linking. BOAMP also has no explicit renewal label, which means the linking output is necessarily an inferred event proxy.

6 Duration Field Reliability

Declared duration is not the survival target. In the `APPEL_OFFRE` population, the cleaned declared duration field is heavily concentrated on administrative values of 12, 24, 36, and 48 months.

Table 4: Declared duration concentration in `APPEL_OFFRE` notices

Duration value	Count	Share of non-missing durations
48 months	462	31.2%
12 months	316	21.4%
24 months	151	10.2%
36 months	99	6.7%
Subtotal exactly 12/24/36/48	1,028	69.5%

This concentration is more consistent with administrative ceilings than with realized renewal timing. For that reason, `declared_duration_months` is used as a covariate and as a temporal search prior, while `observed_duration_months` is the survival time used in Phase 2.

7 Buyer Identification and SIREN/SIRET Enrichment

7.1 Motivation: buyer fragmentation

The baseline pipeline builds a canonical buyer key using a SIRET-first rule: when a valid 14-digit SIRET is present in the BOAMP record, the key is `SIRET:{14-digit}`; otherwise it falls back to a normalized name key `NAME:{normalized_name}` after stripping accents, punctuation, and legal-form suffixes. In the cleaned corpus, approximately 95% of rows carry a name-based key. Because the same public buyer can appear under slightly different spellings across procurements (*e.g.* “Ville de Nantes” vs. “Nantes Métropole”), name-based keys fragment what is in reality a single buyer into multiple groups. Fragmented groups reduce the pool of candidate renewals available to the linker and can produce false negatives in the event variable.

7.2 Data source and tool

Enrichment uses the *API Recherche d’Entreprises* (<https://recherche-entreprises.api.gouv.fr>), the official open-data search interface for the French SIRENE register. It is publicly accessible, requires no authentication, and covers all legal entities registered in France, including public bodies (*communes, départements, établissements publics, hospitals, intercommunalités*). The API returns up to ten candidates per query with a declared rate limit of 7 requests per second; the pipeline uses a conservative limit of 5 requests per second with local caching to avoid redundant calls on re-runs.

7.3 Why SIREN, not SIRET

SIRET (14 digits) identifies a specific establishment of a legal entity. SIREN (9 digits) identifies the legal entity itself. For buyer grouping, SIREN is more appropriate: a single public buyer (*e.g.* a *département*) can have multiple establishments with different SIRETs but a single SIREN. Using SIREN as the enriched key prefix (`SIREN:{9-digit}`) makes the source and grouping level unambiguous and distinguishes enriched keys from BOAMP-sourced keys (which retain the `SIRET:` prefix).

7.4 Matching method and confidence levels

For each unique buyer name, the pipeline:

1. Detects the likely entity type from the name (commune, SDIS, CHU, région, EPCI, etc.) to select the appropriate *nature juridique* API filter and query formulation.
2. Queries the API with the normalized buyer name, filtered by department code when available.
3. Scores every returned candidate using a token-sort string similarity (rapidfuzz) on normalized names, with a small bonus for token overlap and a department-match check.
4. Classifies the best candidate into one of four confidence levels:

Level	Score threshold	Interpretation
HIGH	≥ 85 , or ≥ 75 with single candidate or margin ≥ 10	Strong match; used automatically for <code>buyer_key_enriched</code>
MEDIUM	≥ 70 (not HIGH)	Plausible but ambiguous; retained for manual review only
LOW	≥ 55 (not MEDIUM)	Weak or generic match; not used automatically
NO_MATCH	< 55 or no candidates	No reliable candidate found

Table 5: Confidence classification for SIREN enrichment.

7.5 New buyer key

A new column `buyer_key_enriched` is defined with a simple rule: use `SIREN:{siren}` when the match is HIGH confidence and the SIREN is valid; otherwise keep the original `buyer_key`. MEDIUM, LOW, and NO_MATCH rows are therefore unchanged. The original `buyer_key` is preserved in `buyer_key_original`. Confidence levels MEDIUM and LOW are surfaced in `buyer_siren_enrichment/outputs/ambiguous_buyer_matches_for_review.csv` for optional manual validation but are never applied automatically.

7.6 Coverage and effect on renewal linking

Of the 1,210 eligible contracts, 724 carry a `SIREN:` key (plus 2 `SIRET:`) and 484 a `NAME:` key after enrichment; 275 of 525 unique buyers (52.4%) matched with HIGH confidence. The SIREN-based merges consolidated buyer aliases; post-enrichment eligible linking rate: 55.0% (665/1,210). An **auxiliary Jaccard-based comparison** was also run separately from the official Sentence-Transformer phase-2 pipeline, using `buyer_key_enriched` for grouping (window ± 6 months, Jaccard threshold 0.20, weighting 0.7 text + 0.3 time). It produces two auxiliary files, `boamp_full_survival.csv` (name-based baseline, 1,933 rows) and `boamp_full_survival_enriched.csv` (SIREN-enriched, 1,933 rows) in `boamp_renewal_linking_siren_enriched/outputs/`; these are **legacy comparison artifacts**, not the current-stage survival dataset. Numerical comparison results are saved in `buyer_siren_enrichment/outputs/baseline_vs_siren_enriched_linking_comparison.csv`.

7.7 Limitations

- **API results are not ground truth.** The SIRENE register records the entity as legally registered, not necessarily under the name used in procurement. Fuzzy matching can produce false positives, particularly for common generic names.
- **MEDIUM and LOW matches are not applied automatically.** They are kept in the review file for optional human validation.
- **HIGH-confidence mistakes remain possible.** A score ≥ 85 does not guarantee a correct match; it reflects strong textual similarity under a specific normalization scheme. Rare false positives at HIGH confidence would introduce a small number of incorrect group merges.
- **NO_MATCH buyers are unresolved.** Buyers with acronyms, internal codes, or abbreviated names (*e.g.* “CNIEG”, “SAELSI”, “DIR Ouest”) often fail to match and remain name-keyed. Their renewal groups are unchanged relative to the baseline.
- **The SIREN enrichment annotation does not recompute the event variable.** The official phase-2 dataset (`boamp_phase2_survival.csv`, 665 events) incorporates SIREN-based buyer key merges into its `buyer_key` column (724 SIREN: keys, 484 NAME: keys, 2 SIRET: keys). The file `boamp_phase2_survival_siren_enriched.csv` extends it with four annotation columns (`buyer_key_enriched`, `enriched_siren`, `enrichment_confidence`, `enrichment_score`) added by left-join; these record enrichment metadata but do not change the event or survival-time columns.

Scientific framing. SIREN enrichment is a buyer-resolution improvement, not a new ground truth. It is expected to reduce false negatives arising from buyer fragmentation and to improve the precision of renewal-group boundaries, but it cannot compensate for the fundamental limitation that **event** remains a proxy.

8 Construction of the event Variable

8.1 Exact definition

event = 1 is assigned when the linking algorithm finds a later BOAMP APPEL_OFFRE notice from the same buyer within a window of [6, 72] months around the source contract’s expected end date, passing a hard `text_similarity` ≥ 0.20 filter, and achieving the highest composite score among all candidates considered. The renewal variable is constructed by matching later notices from the same `buyer_key` using text, CPV, and timing; BOAMP provides no explicit renewal ID.

8.2 Scoring formula

For source contract i and candidate j the composite score is:

$$S(i, j) = 0.40 s_{\text{text}} + 0.25 s_{\text{cpv}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}}$$

When CPV is missing for either notice, the CPV weight is redistributed proportionally: $S = (0.40 s_{\text{text}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}}) / 0.75$.

8.3 Hard threshold and justification

The **only hard acceptance threshold** in the baseline model is `text_similarity` ≥ 0.20 ; no composite floor is applied. The hard filter is placed on text similarity because semantic similarity of the contract description is the most reliable single discriminator between a genuine renewal and an unrelated later notice from the same buyer. The observed minimum composite score for accepted links is approximately 0.37, arising from the weighting formula applied to pairs that pass the text filter. Two post-hoc confidence tiers are defined for sensitivity analysis:

- Medium/high confidence: composite ≥ 0.50 — 400 of 665 events (60.2%);
- Strict high-confidence: composite ≥ 0.70 *and* score margin ≥ 0.05 , multi-candidate links only — 75 events (11.3%).

8.4 Interpretation of `event = 0`

`event = 0` is right-censored: no successor was detected under the rule before the 2024-12-31 study end. Right-censoring does *not* mean the contract was not renewed; it means no identifiable BOAMP successor was found under the current chaining rule (for example, the renewal may fall after study end, may be procured through a channel that does not generate a new BOAMP notice, or may use a buyer name that fragments away from the source key). The variable is therefore a **proxy chaining result**, not a legally certified renewal and not a confirmed absence of renewal.

9 Validation and Robustness of the event Variable

9.1 Confidence tiers

The confidence diagnostics are useful for sensitivity analysis, not as legal ground truth. A strict subset can bound the event definition, but it does not prove that the remaining links are false.

Table 6: Linking confidence summary

Metric	Value	Interpretation
Linked events	665	Proxy renewal events retained in the official BOAMP hand-off.
Right-censored	545	No observed link before study end under the chosen rule.
Linking rate	55.0%	Share of eligible contracts with <code>event = 1</code> .
Strict high-confidence links	75	Composite score ≥ 0.70 and score margin ≥ 0.05 , <i>excluding</i> the 164 single-candidate links (margin undefined). A broader count of 86 results when single-candidate links with score ≥ 0.70 are also included.
Single-candidate links	164	Only one surviving candidate after filtering; margin is undefined for these.
Ambiguous multi-candidate links	219 of 501	43.7% of multi-candidate events have a score margin below 0.05.
Main unlinked failure mode	491 of 545	Most unlinked contracts have no temporal partner in the search window.
Secondary failure modes	25 text, 29 CPV	Some eligible contracts had a temporal partner but failed semantic or CPV checks.

9.2 Threshold sensitivity

Table 7 reports survival model outcomes under seven increasingly strict event definitions, sourced from `validation_robustness/outputs/threshold_sensitivity_summary.csv`.

Table 7: Threshold sensitivity of the survival model. KM median in months; “—” = median not reached within the study window.

Scenario	Events	Rate	KM med.	Cox C
Baseline (all accepted links)	665	55.0%	50.1	0.632
Score ≥ 0.50	400	33.1%	—	0.616
Score ≥ 0.60	205	16.9%	—	0.599
Score ≥ 0.70	99	8.2%	—	0.473
Score ≥ 0.80	54	4.5%	—	0.553
Score ≥ 0.90	18	1.5%	—	—
Strict HC ($S \geq 0.70$, $\Delta \geq 0.05$)	86	7.1%	—	0.488

The absolute event rate and KM survival level are sensitive to the event definition. The Cox concordance ranges from 0.632 (baseline) to 0.473 (score ≥ 0.70), declining more sharply at high thresholds than in the W=12 run; the baseline and score ≥ 0.50 tier (0.616) remain useful. The extreme ≥ 0.90 cut (18 events) is too sparse for stable estimates. The direction of both significant hazard ratios (HR < 1 for declared duration, HR > 1 for start year) is preserved in every scenario. The qualitative conclusions of the survival model are therefore robust to the choice of linking threshold.

9.3 Uncertain cases and score margins

The median composite score of the 665 accepted links is 0.529 (IQR 0.453 – 0.626). Of the 501 multi-candidate links, 219 (43.7%) have a score margin below 0.05 — the algorithm selected a winner but an alternative was nearly equally plausible. Removing all uncertain links (composite < 0.50 or margin < 0.05) reduces the analysis to 942 rows and 397 events (42.1%); KM median is not reached, and the Cox concordance changes from 0.6317 to 0.579. The drop in concordance confirms that the uncertain links carry positive discriminative signal and do not appear to introduce dominant systematic bias.

9.4 Manual validation (executed 2026-07-02)

A stratified 150-row sample was drawn and the record-level review has been **executed** against the full official BOAMP notice records (labeled file: `event_validation/outputs/manual_validation_audit_labeled.csv`; summary: `event_validation/manual_validation_summary.md`).

Table 8: Manual validation sample design (150 rows). Review executed 2026-07-02.

Stratum	Criterion	<i>n</i>
A — High confidence	<code>event=1</code> , strict HC (<code>high_confidence_strict</code>)	20
B — Ambiguous margin	<code>event=1</code> , margin < 0.05 , multi-candidate	20
C — Single-candidate	<code>event=1</code> , <code>single_candidate_match=True</code>	15
D — No successor	<code>event=0</code>	50
E — Remaining linked	<code>event=1</code> , medium/low confidence, not in A–C	45
Total		150

Results. On the 100 audited linked pairs: 14 credible renewals, 82 false positives, 4 uncertain — raw precision 0.146, or ≈ 0.088 when the strata are re-weighted to the 665-event population. Precision is strongly graded by signal quality: 0.50 in the HIGH tier and under the strict flag, 1.00 at text

similarity ≥ 0.80 , 0.09 in the MEDIUM tier, and 0.00 in the LOW tier. On the 50 audited unlinked sources: 41 plausible censorings, 6 missed renewals (12.8% of decided cases), 3 uncertain — the dominant miss causes are early renewals published before the ± 6 -month window opens and buyer renames. These are *sample-based audit estimates*, not population-level ground truth, and the “true renewal” judgment itself is based on notice content, not legal records. The practical conclusion: the event variable is not validated at the baseline definition; only its high-text-similarity core carries reliable renewal meaning, and a re-linking with a higher text floor, corrected generic-CPV scoring and a wider window is required before the survival results can be interpreted as renewal timing.

9.5 Event/censoring bias comparison

Covariate balance between linked (event= 1) and censored (event= 0) contracts is good on all survival-model predictors: the standardised mean difference (SMD) is 0.166 for declared duration, -0.150 for start year, and -0.010 for the imputation flag — all below the conventional 0.20 threshold for substantial imbalance. The largest observed imbalance is amount non-missing ($\Delta = -0.084$): censored contracts are more likely to carry a reported amount, possibly because high-value contracts are renegotiated through channels that do not produce a new BOAMP notice. This does not affect the survival model (amount is not a covariate).

Event rate by cohort year is broadly stable at 51–67% for contracts starting in 2015–2020 and 2022 (2021: 41%). The 2023 cohort shows only 8 linked events out of 33 eligible (24.2%), but the sample is small and hard to interpret as a trend; the short observation window (less than 24 months to study end) is the more likely explanation. The dominant source of bias remains **structural right-censoring** of the 2022–2024 cohort, whose expected renewal windows fall after the 2024-12-31 study end — a study-design constraint, not a measurement error in the linking rule.

In summary, the 55.0% linking rate should be read as a **chaining result**: the proportion of eligible contracts for which the algorithm identified a plausible BOAMP successor under the current rule. It is *not* a measurement of the true renewal rate: the executed manual audit estimates that only ≈ 9 –15% of the accepted links are credible renewals (concentrated at high text similarity), while $\approx 13\%$ of audited censored contracts hide a missed renewal. The sensitivity analysis shows the qualitative covariate conclusions are stable across threshold scenarios, but the audited precision means the variable is at best a *weak* prioritisation signal at the baseline definition; it approaches renewal meaning only in its high-text-similarity core, provided its proxy nature is communicated wherever the model output is used.

10 Limitations and Impact on Survival Analysis

Selection bias remains possible because contracts with no temporal partner cannot be linked even if a real-world renewal occurred outside the observable window. Buyer fragmentation can create false negatives, while generic or repetitive contract descriptions can create false positives. Missing or generic CPV reduces auditability. Contracts from late 2023 face compressed observation windows, and 2024 contracts are structurally excluded from the eligible denominator because their expected renewal window falls after study end. The executed manual audit estimates the precision of the proxy renewal event at ≈ 9 –15% overall (0.50 in the HIGH/strict tier), with a 12.8% missed-renewal rate among audited censored contracts.

Table 9: Principal data quality risks

Risk	Evidence in current data	Modeling impact
Buyer fragmentation	Post SIREN enrichment, 724 of 1,210 eligible rows carry a SIREN: key (plus 2 SIRET:) and 484 a NAME: key. Residual name-based keys may still fragment some buyers.	Residual structural false negatives when the same buyer appears under multiple name spellings not resolved by the SIREN API.
Generic or missing CPV	Raw CPV missingness is 1.9%; generic codes are capped at low similarity.	Lower match precision and weaker audit trail for some links.
Sparse amount field	Only 21.4% of source APPEL_OFFRE notices have a usable amount.	Amount cannot serve as a reliable matching signal or segmentation variable.
Administrative duration values	69.5% of observed source durations equal 12, 24, 36, or 48 months.	Declared duration should not be interpreted as realized renewal time.
Observation-window censoring	545 eligible contracts are right-censored; late cohorts have less time to reveal renewals.	Apparent non-renewal may reflect study-end truncation rather than true absence of renewal.
Proxy event definition	BOAMP contains no explicit renewal label.	Event quality depends on the linking rule and should be stress-tested with confidence diagnostics.

11 Reproducibility

This report is reproducible from tracked project outputs. The core files are:

- `data/processed/boamp_full_clean.csv`
- `data/processed/boamp_phase2_survival.csv`
- `boamp_renewal_linking_quality/outputs/boamp_linking_stats.csv`
- `boamp_renewal_linking_quality/outputs/boamp_renewal_links.csv`
- `boamp_renewal_linking_quality/outputs/boamp_renewal_candidates.csv`
- `boamp_renewal_linking_quality/outputs/boamp_bias_report.csv`
- `boamp_renewal_linking_quality/outputs/boamp_link_confidence_diagnostics.csv`

The relevant pipeline steps are implemented in `scripts/task_boamp_full_clean.py`, `scripts/task9_boamp_phase2_handoff.py`, and `scripts/task10_link_confidence_diagnostics.py`. No manual, untracked data manipulation is required to reproduce the counts reported here.

12 Conclusion

The BOAMP dataset satisfies the internship guide’s minimum data-quality requirement for Phase 1 because it provides a documented corpus, explicit cleaning rules, missingness information, a transparent linking rate, and a survival-ready population of 1,210 eligible contracts with either observed or censored duration. The main limitation is conceptual rather than technical: **event** remains a proxy renewal event, not legal ground truth. The 55.0% figure is a chaining/linking rate, not a measured true renewal rate; SIREN enrichment is a buyer-resolution improvement, not ground truth. The 150-row manual review (executed 2026-07-02) puts an empirical bound on event quality: precision $\approx 9\text{--}15\%$ overall, 0.50 in the HIGH/strict tier, 1.00 at text similarity ≥ 0.80 , and a 12.8% missed-renewal rate among audited censored contracts. Re-linking under stricter rules and re-estimating the survival models is the priority next step.